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SOFTWARE REQUIREMENT SPECIFICATION

AETHERIS

DIGITAL ENVIRONMENT TWIN SYSTEM

Prepared in IEEE Std 830-1998 style

Under the Guidance of

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Document Status and Interpretation

This Software Requirements Specification (SRS) document has been prepared for **AETHERIS — Digital Environment Twin System**, based on the project report, system architecture, implemented features, and available test results. It follows the recommended structure of IEEE Std 830-1998 and provides a formal description of the software requirements, interfaces, database design, operating environment, and system constraints.

The primary purpose of this document is to establish a clear understanding of the system's expected functionality among developers, testers, faculty evaluators, and future maintainers. It serves as a reference for software development, testing, verification, and project evaluation, ensuring consistency between the proposed requirements and the implemented application.

The document includes technical diagrams such as the System Context Diagram, Use Case Diagram, Data Flow Diagrams (DFDs), Flowcharts, Sequence Diagrams, UI Wireframes, Network Topology, and Entity–Relationship Diagram (ERD). These diagrams illustrate the system architecture, user interactions, data processing workflows, communication between components, and database relationships.

AETHERIS is an academic prototype that provides environmental data visualization, Clean Air Score calculation, GPS-based nearest-station identification, hazard simulation, and environmental alert generation. The current implementation uses simulated telemetry by default. Performance targets that have not been experimentally verified are identified as proposed acceptance criteria and require validation under the intended deployment environment.

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, interface, data, performance, security, reliability, maintainability and availability requirements for AETHERIS, a full-stack Digital Environment Twin System. It is intended for the project team, faculty evaluators, developers, testers, database designers and future maintainers.

The specification provides a common baseline for implementation, verification and acceptance. It covers the product showcase, citizen/business dashboard, engineering console, geolocation pairing, telemetry visualization, hazard simulation, alert handling and environmental audit report generation.

1.2 Scope

AETHERIS presents a virtual representation of environmental sensor stations and their telemetry. It displays air quality and microclimate metrics, translates AQI into a Clean Air Score and citizen-friendly verdict, estimates the nearest station from browser location, simulates defined hazard scenarios, raises threshold alerts and provides a print/PDF-style environmental audit report.

Included in Scope	Explicitly Out of Scope for the Current Version
Account registration, login, token-based access and role-based permissions.	A production-grade identity provider, SSO, MFA or enterprise directory integration.
Station registration and station detail/list operations.	Automatic discovery/provisioning of physical IoT hardware.
Live and historical telemetry views, with scheduled simulated variation.	Guaranteed real-time measurements from physical sensors; actual hardware integration is future work.
2.5D HTML5 Canvas twin, particulate dispersion, wind and thermal visualization modes.	High-fidelity computational fluid dynamics or scientifically validated atmospheric forecasting.

GPS-based nearest-station pairing, distance, bearing and compass label.	Guaranteed GPS accuracy or background tracking without user permission.
HEATWAVE, INDUSTRIAL_EMISSION, RAIN_CLEANSING and NORMAL scenarios; alerts and acknowledgement.	Unbounded user-defined scenarios or direct control of industrial equipment.
Clean Air Score, citizen verdicts, guided tour and audit-style report output.	Legal certification, regulatory approval or a guarantee of ISO 14001 compliance.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
AETHERIS	Digital Environment Twin System described by this SRS.
AQI	Air Quality Index; a summarized indicator of air pollution.
Clean Air Score	A 0–100 score calculated from AQI, where a higher score indicates cleaner conditions.
Digital Twin	A virtual representation of an environment or its sensor stations, updated with telemetry or simulation.
DTO	Data Transfer Object used to transfer structured data across application layers.
GPS	Global Positioning System used by the browser to request device coordinates.
H2	In-memory relational database used as a fallback when MySQL is unavailable at startup.
JPA	Java Persistence API; object-relational persistence abstraction.
PM2.5 / PM10	Particulate matter with aerodynamic diameters up to approximately 2.5 / 10 micrometres.
RBAC	Role-Based Access Control.
REST API	HTTP interface organized around resources and operations.

Telemetry	Environmental measurements associated with a station and timestamp.
UI	User Interface.
CORS	Cross-Origin Resource Sharing policy controlling browser cross-origin requests.

1.4 Overview

Section 2 describes the product context, capabilities, user classes, operating environment, constraints and assumptions. Section 3 specifies external interfaces, functional requirements with unique identifiers, measurable performance expectations, database requirements and software quality attributes. Section 4 contains supporting assumptions, interface/API summaries, acceptance checks and references.

2. Overall Description

2.1 Product Perspective

AETHERIS is a browser-accessible web application built around a four-layer architecture: presentation, REST controller, business logic and data persistence. The Spring Boot application hosts the static frontend and exposes REST endpoints, normally on port 9090. The frontend polls telemetry endpoints and renders the dashboard and interactive Canvas twin. MySQL is the intended persistent store; an in-memory H2 database is used when MySQL is unavailable at startup.

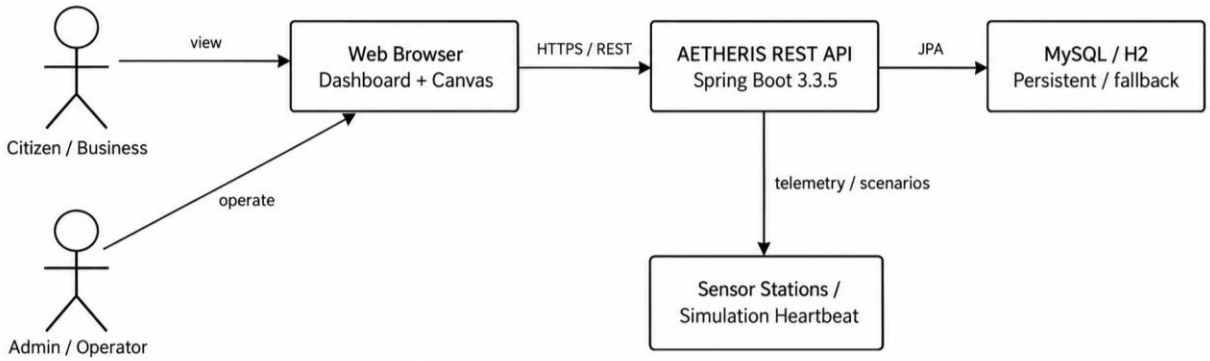


Figure 2.1: System Context Diagram

External dependencies include the user's browser, browser geolocation permission/service, the configured database, and—when real measurements are desired—future sensor hardware or a gateway. In the documented prototype, the scheduled simulation heartbeat generates small telemetry variations.

2.2 Product Features

- Product showcase landing page and guided product tour.
- User registration, login by username or e-mail, profile retrieval and role-based access control.

- Citizen/business dashboard with Clean Air Score, AQI verdict and six metric cards: Air Purity, Real-Feel Temperature, Solar UV, Natural Breeze, Acoustic Serenity and Atmospheric Freshness.
- Interactive 2.5D Canvas twin with wind currents, PM2.5/PM10 dispersion and thermal gradients across three visualization modes.
- Station list/detail/registration and nearest-station pairing using Haversine distance, bearing and an eight-point compass label.
- Live telemetry polling and historical time-series retrieval.
- What-if simulation for HEATWAVE, INDUSTRIAL_EMISSION, RAIN_CLEANSING and NORMAL.

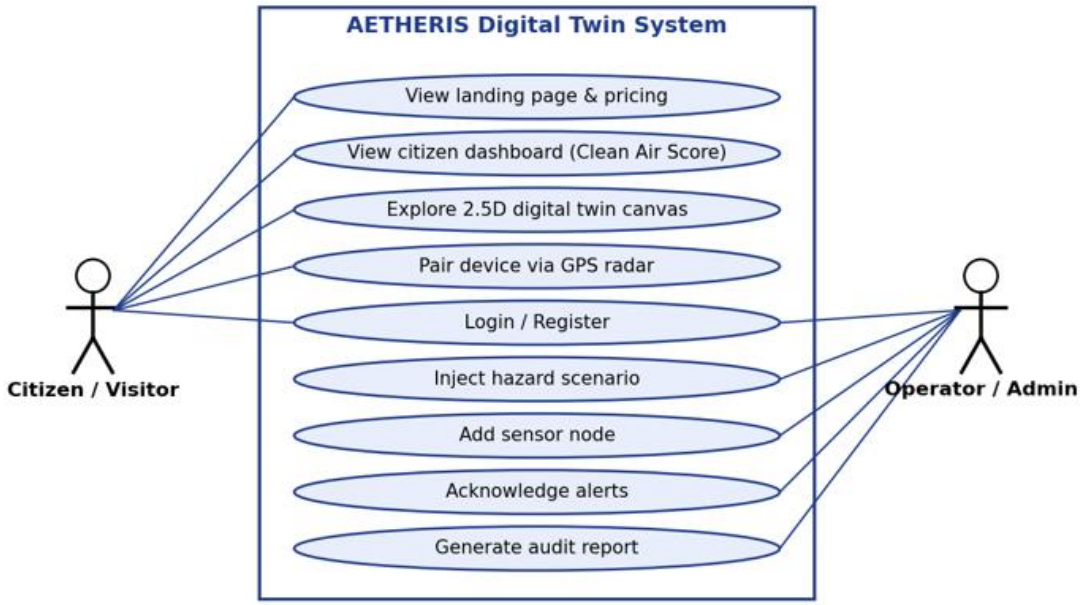


Figure 2.2: Use Case Diagram

2.3 User Classes and Characteristics

User Class	Typical Goals	Permissions / Characteristics
Citizen / Business User	Understand local environmental conditions and health-oriented AQI guidance; optionally pair device to nearest station.	Read dashboard and public-facing telemetry; location is requested only through browser permission.

Researcher	Inspect current and historical telemetry and reports for analysis.	Read-oriented access to live telemetry, history and reports.
Operator	Run authorized what-if scenarios and acknowledge alerts.	Authenticated role; may trigger simulations and acknowledge alerts.
Administrator	Manage station nodes, run scenarios and acknowledge alerts.	Full administrative control, including station registration and protected operations.
Maintainer / Developer	Deploy, configure, troubleshoot and extend the system.	Operational access to application configuration, logs, schema and code; not a separate end-user role in the application.

2.4 Operating Environment

Component	Requirement / Environment
Server runtime	Java 17; Spring Boot 3.3.5 application.
Build	Maven / Maven Wrapper.
Application port	9090 by default; configurable via server.port.
Database	MySQL for persistence; H2 in-memory fallback if MySQL is unavailable at startup.
Client	Modern desktop or mobile browser supporting HTML5 Canvas, JavaScript, Fetch/REST and Geolocation API.
Network	IP network with HTTP/HTTPS connectivity between browser and application; JSON REST messages.
Hardware	General-purpose computer/server capable of running Java 17; graphics-capable browser recommended for Canvas animation.
Sensor source	Scheduled simulated telemetry in the current prototype; real sensor gateway integration is not included.

2.5 Design and Implementation Constraints

- The backend uses Java 17, Spring Boot 3.3.5, Spring Data JPA and REST controllers.
- The frontend uses HTML5, CSS3, JavaScript and HTML5 Canvas; it calls the REST API.
- Persistence is relational and organized around users, stations, telemetry_records and environmental_alerts.
- The application uses salted SHA-256 password hashing and token authentication in the documented prototype. This is an academic implementation; production deployment should replace SHA-256 password hashing with bcrypt or Argon2 and review token lifecycle/security.
- Browser geolocation requires explicit permission and valid coordinates; state-wise preset locations are used when permission is denied.
- Only the four documented scenario values are accepted by the hazard injection engine.
- H2 fallback is volatile and loses data when the application stops; it is not a substitute for production persistence.
- The audit report is audit-style output and must not be represented as external certification or legal compliance approval.

2.6 Assumptions and Dependencies

- The browser supports the required Canvas and geolocation features; users may deny location access.
- At least one station must be registered for nearest-station pairing to return a result.
- MySQL is expected to be available for durable storage; H2 fallback is intended for graceful startup/demo operation.
- Telemetry may be simulated; system users must be told whether displayed readings are simulated or sensor-derived.
- The user's device and network influence GPS precision and dashboard refresh latency.
- Threshold values for environmental alerts are application configuration/business rules and should be explicitly documented for each deployment.
- Performance targets in Section 3.3 are proposed acceptance targets, not measured guarantees from the supplied report.

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interface

The UI shall provide a product landing page, login/register modal, citizen/business dashboard, 2.5D Digital Twin Canvas, GPS pairing radar, hazard injection panel, alert view and environmental audit report. Visual components should communicate units and data freshness, and distinguish simulated telemetry from readings received from connected hardware.

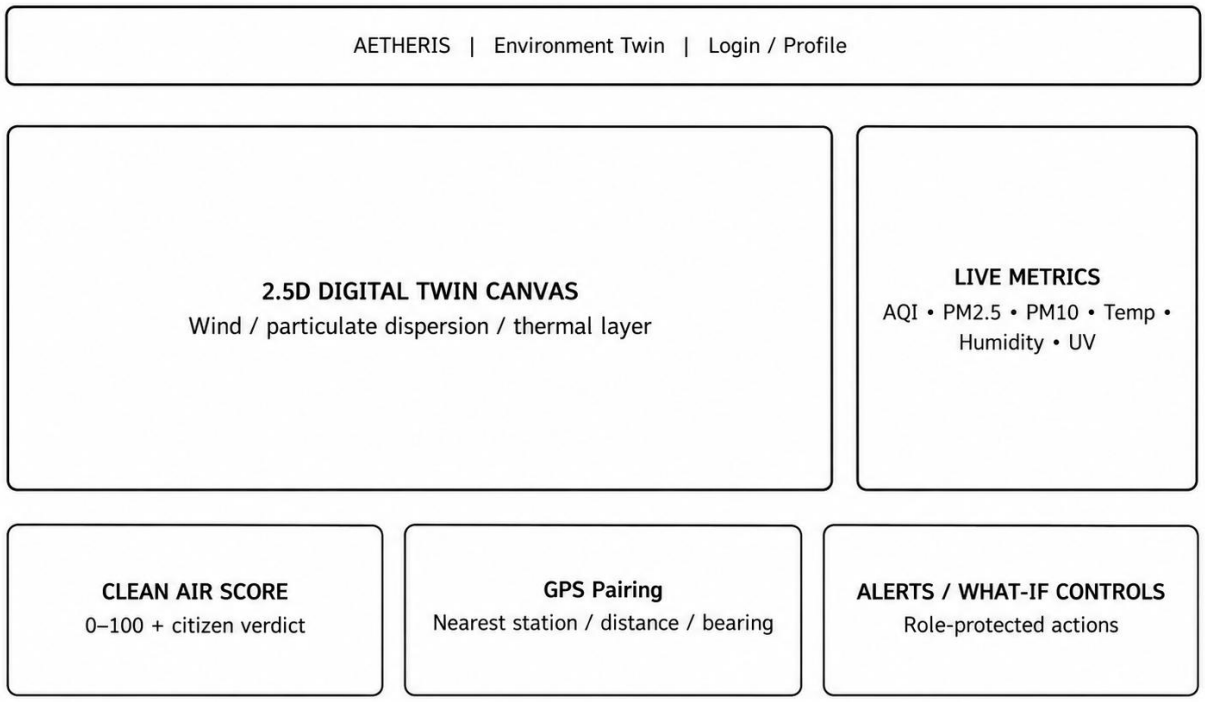


Figure 3.1: Conceptual UI Wireframe

UI Screen / Component	Required Behaviour
Landing page	Explain the AETHERIS purpose and provide entry points to the dashboard and guided tour.
Authentication modal	Accept username/e-mail and password for login; collect required fields during registration; display validation failures.
Citizen dashboard	Show Clean Air Score (0–100), citizen verdict, metric cards and health-oriented advice.

Digital Twin Canvas	Visualize stations and selected environmental mode; show wind/particulate/thermal effects.
GPS pairing radar	Ask for location permission; display nearest station, distance, bearing and compass direction, or fallback state preset.
Engineering controls	Expose station registration, scenario injection and alert acknowledgement only to authorized roles.
Audit report	Present environmental summary in a print-friendly layout and support browser print/PDF workflow.

3.1.2 Hardware Interfaces

No direct hardware interface is required for the current simulated-data prototype. The application shall support station metadata and telemetry through its REST API. Physical sensors, IoT gateways and hardware protocols such as MQTT are future integrations and are not assumed to exist in the current build.

3.1.3 Software Interfaces

Interface	Specification
Browser frontend → backend	REST over HTTP(S), JSON payloads, configurable application base URL.
Backend → database	Spring Data JPA / relational persistence; MySQL primary and H2 fallback.
Browser → Geolocation API	Browser-native location request subject to user permission.
Backend scheduler	Spring @Scheduled heartbeat updates simulated station telemetry.
Canvas rendering	HTML5 Canvas API for 2.5D scene, particles and environmental layers.
Build/runtime	Maven Wrapper and Java 17.

3.1.4 Communication Links and Network Topology

The browser communicates with the Spring Boot application through REST endpoints. The backend accesses MySQL or the H2 fallback through JPA. CORS is enabled for a standalone

frontend deployment; allowed origins should be restricted in production. Production deployment should use HTTPS and should not expose database ports to public clients.

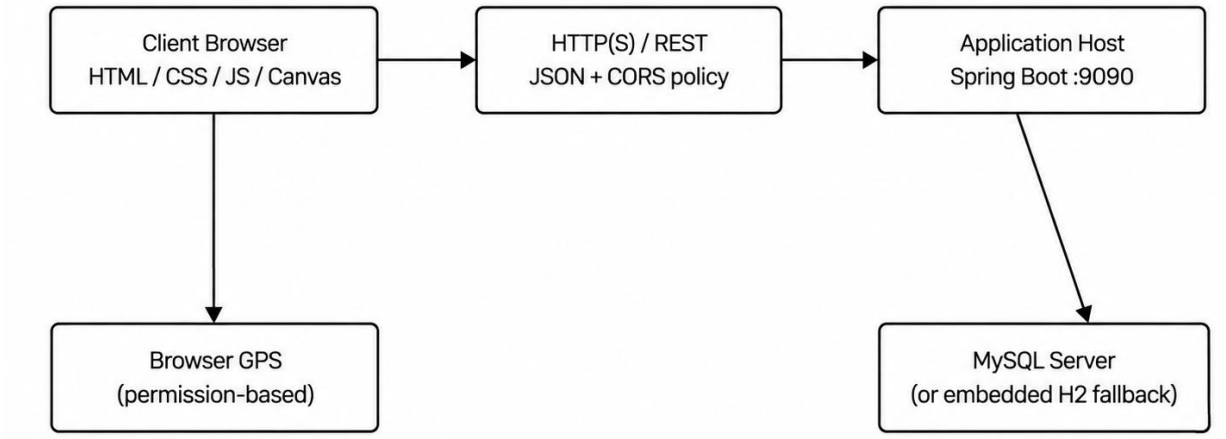


Figure 3.2: Deployment and Network Topology

3.1.5 REST API Interface

Method	Endpoint	Purpose / Expected Result
POST	/api/auth/login	Authenticate user; return token on success.
POST	/api/auth/register	Register a new user account; reject duplicate username/e-mail.
GET	/api/auth/me	Return authenticated user profile.
POST	/api/auth/update-location	Update stored user device coordinates.
GET	/api/stations	List registered sensor stations.
GET	/api/stations/{id}	Return station details by identifier.
POST	/api/stations	Register a physical or virtual sensor node; role protected.
GET	/api/telemetry/live	Return current telemetry across stations.
POST	/api/simulation/trigger	Trigger one supported scenario; role protected.
GET	/api/alerts	List alerts; supports unacknowledgedOnly=true.
POST	/api/alerts/{id}/acknowledge	Acknowledge an alert; role protected.

3.2 Functional Requirements

The following functional requirements define the expected behaviour of the AETHERIS Digital Environment Twin System. Each requirement is assigned a unique identifier to support traceability, implementation, and testing.

3.2.1 Authentication and Authorization

FR-AUTH-01 — User Registration: The system shall allow users to register using a username, e-mail address, and password. It shall validate required fields, prevent duplicate usernames and e-mail addresses, and securely store password credentials.

FR-AUTH-02 — User Login: The system shall authenticate users using their username or e-mail address and password. Upon successful authentication, it shall issue an authentication token; invalid credentials shall be rejected.

FR-AUTH-03 — Role-Based Authorization: The system shall enforce role-based access for ADMIN, OPERATOR, and RESEARCHER users. Administrators shall manage stations, trigger simulations, and acknowledge alerts; operators shall trigger simulations and acknowledge alerts; researchers shall access authorized telemetry, historical data, and reports.

FR-AUTH-04 — User Profile and Location: The system shall provide the authenticated user's profile and support location updates using valid latitude and longitude coordinates.

3.2.2 Stations and Geolocation

FR-GEO-01 — Station Catalogue: The system shall display registered stations and retrieve station details by their unique identifiers, returning an appropriate error for unknown stations.

FR-GEO-02 — Station Registration: The system shall allow authorized administrators to register sensor stations with details such as name, geographic coordinates, elevation, and status. Latitude and longitude shall be validated against their permissible geographic ranges.

FR-GEO-03 — Nearest-Station Pairing: The system shall identify the nearest registered station using the Haversine formula and provide its details, distance in kilometres, bearing, and corresponding compass direction.

FR-GEO-04 — Location Fallback: If browser location permission is denied, the system shall use the configured state-wise preset location. If no stations are registered, it shall display an appropriate message.

3.2.3 Telemetry and Dashboard

FR-TEL-01 — Live Telemetry: The system shall retrieve and display current environmental telemetry, including available measurements and timestamps, while handling empty datasets appropriately.

FR-TEL-02 — Historical Telemetry: The system shall retrieve historical telemetry records for a selected station in chronological order for visualization and analysis.

FR-TEL-03 — Simulated Telemetry Updates: The system shall periodically update simulated environmental measurements through a scheduled background process and distinguish simulated data from actual sensor readings.

FR-DASH-01 — Clean Air Score: The system shall calculate a Clean Air Score between 0 and 100 using the formula:

Clean Air Score = round (100 − AQI / 5)

The AQI value shall be bounded between 0 and 500 before calculation.

FR-DASH-02 — Citizen Verdict: The system shall classify AQI values into the project's defined categories: Crisp & Pristine (0–50), Moderate (51–100), Sensitive Warning (101–200), and Hazardous (above 200), and display corresponding guidance.

FR-DASH-03 — Temperature Conversion: The system shall convert temperature from Celsius to Fahrenheit using the formula °F = (°C × 9/5) + 32.

FR-DASH-04 — Environmental Visualization: The system shall display environmental information through an interactive 2.5D Canvas digital twin, supporting wind, particulate dispersion, and thermal visualization modes.

3.2.4 Hazard Simulation and Alerts

FR-SIM-01 — Scenario Validation: The system shall accept only the supported scenarios HEATWAVE, INDUSTRIAL_EMISSION, RAIN_CLEANSING, and NORMAL. Scenario execution shall require ADMIN or OPERATOR authorization.

FR-SIM-02 — Scenario Simulation: The system shall apply the following effects to the simulated environment:

- **HEATWAVE:** Increase temperature and ozone levels.
- **INDUSTRIAL_EMISSION:** Increase PM2.5 and CO_2 levels.
- **RAIN_CLEANSING:** Reduce particulate levels and cool the environment.
- **NORMAL:** Restore the nominal baseline.

FR-ALERT-01 — Alert Generation: The system shall generate and store an environmental alert when configured telemetry thresholds are crossed, recording the station, severity, message, acknowledgement status, and creation time.

FR-ALERT-02 — Alert Retrieval: The system shall retrieve environmental alerts and support filtering to display only unacknowledged alerts.

FR-ALERT-03 — Alert Acknowledgement: The system shall allow authorized users to acknowledge existing alerts and update their acknowledgement status.

3.2.5 Reports and Guided Tour

FR-REP-01 — Environmental Report: The system shall generate a print-friendly environmental summary containing relevant station information, telemetry, alerts, and a report timestamp. The report shall support browser printing or saving as PDF.

FR-TOUR-01 — Guided Product Tour: The system shall provide a guided tour introducing users to the landing page, dashboard, and principal digital twin features.

3.2.6 Process Diagrams



Figure 3.3: Data Flow Diagram — Level 0 (Context)

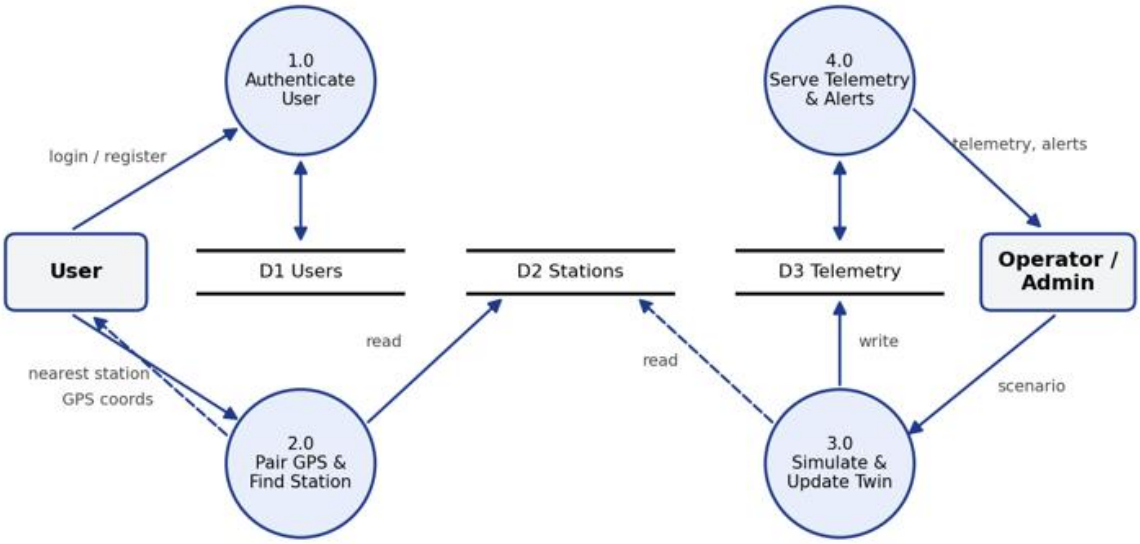


Figure 3.4: Data Flow Diagram — Level 1 (Major Processes)

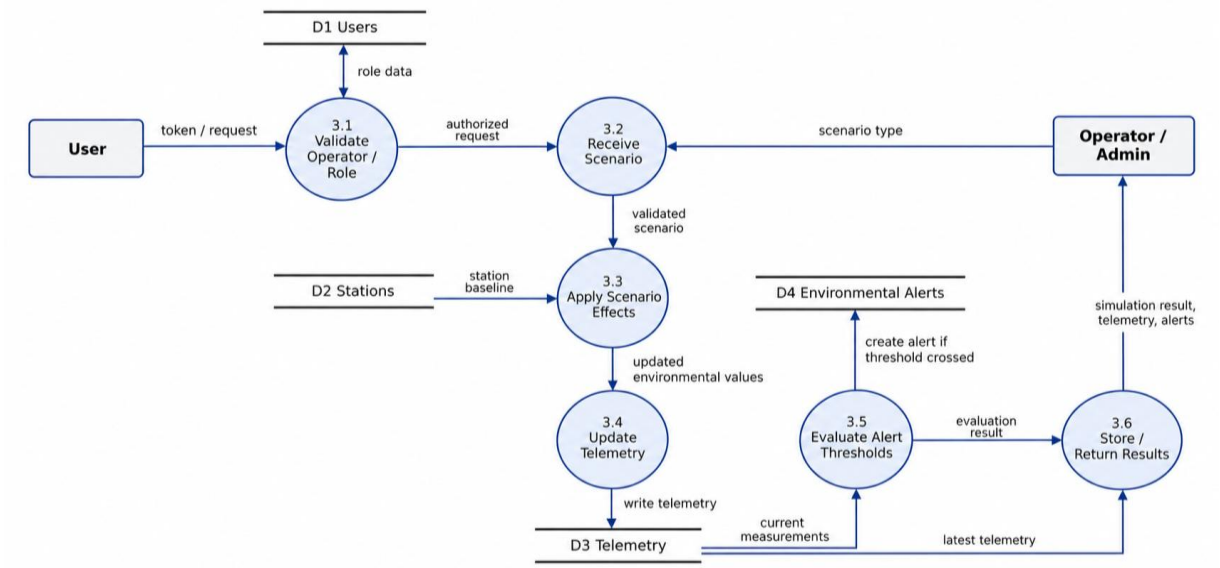


Figure 3.5: Data Flow Diagram — Level 2

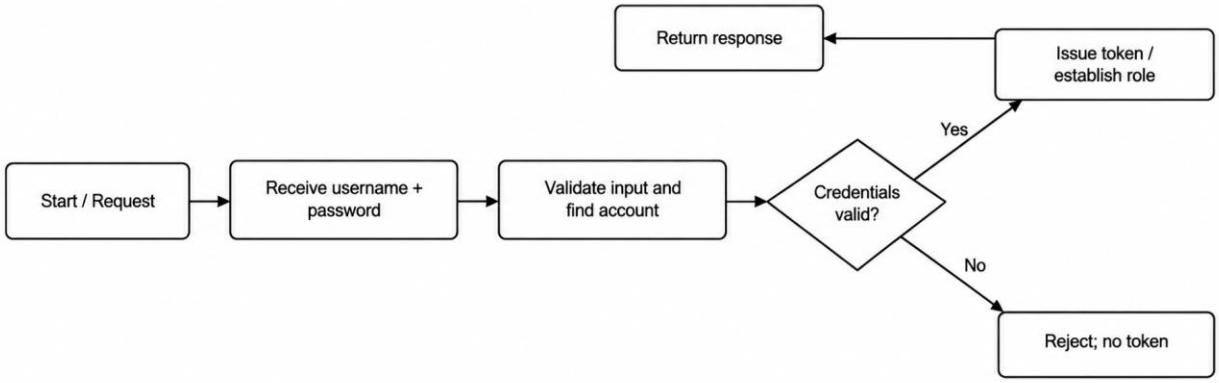


Figure 3.6: Flowchart — Login and Authorization

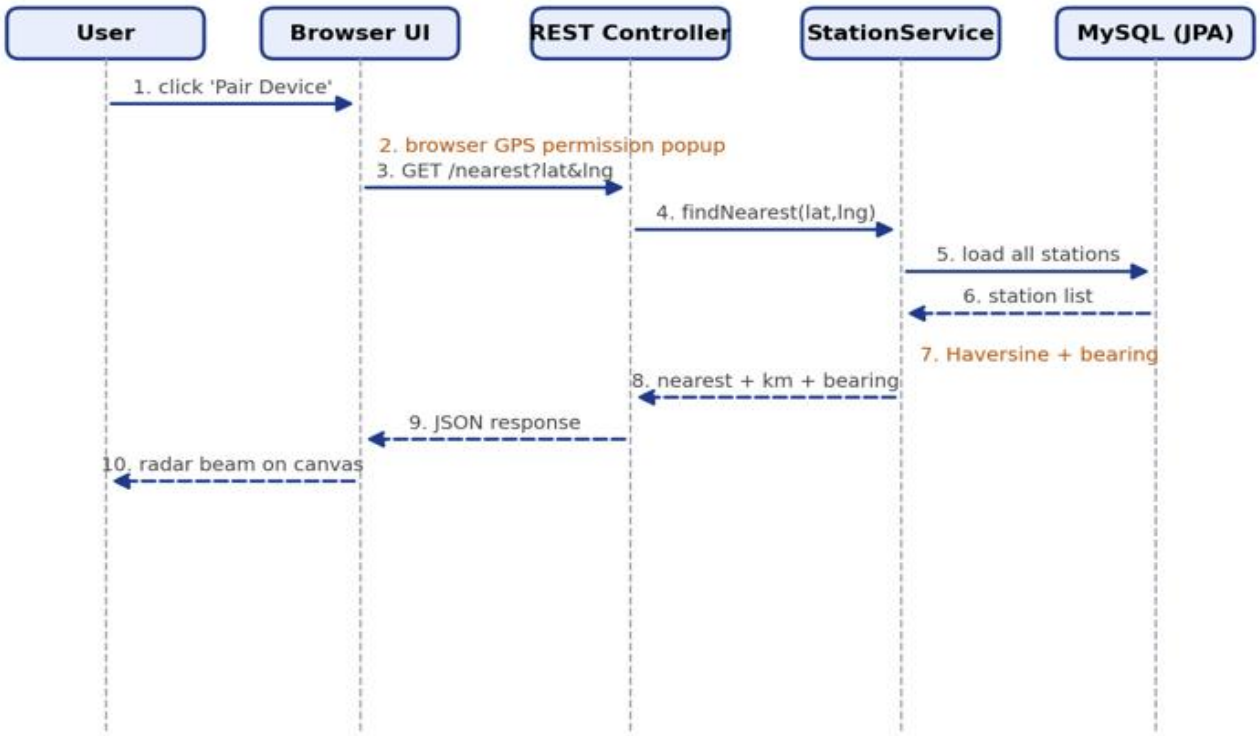


Figure 3.7: Sequence Diagram — GPS Pairing of a Device

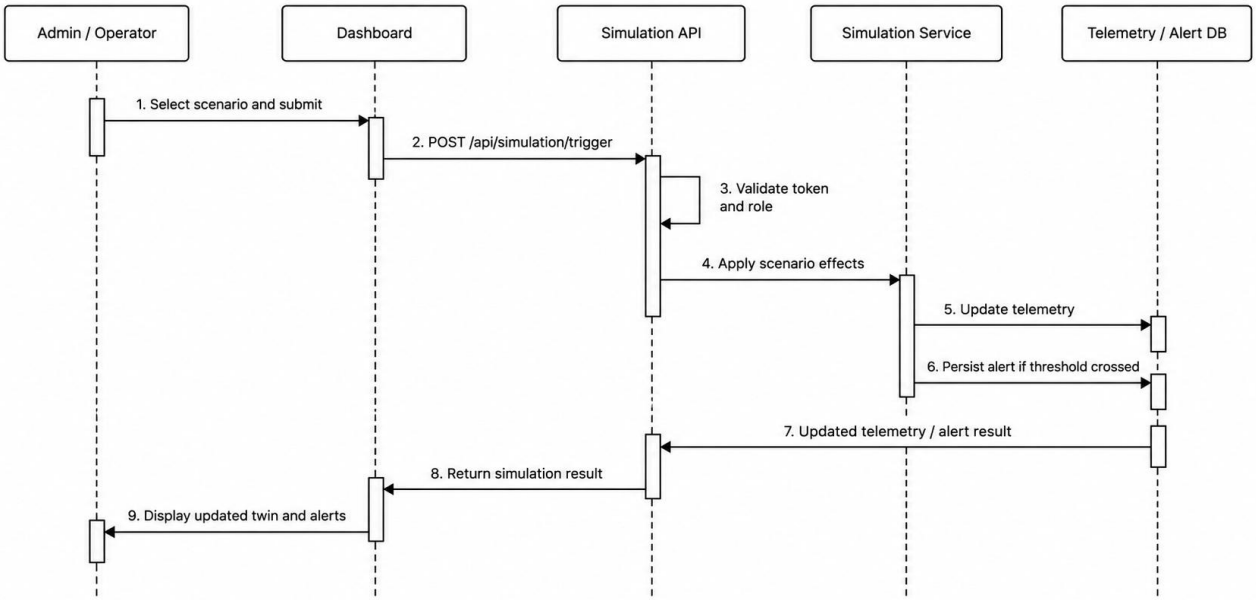


Figure 3.8: Sequence Diagram — Hazard Injection and Alert Generation

3.3 Performance Requirements

The source project report validates selected algorithmic outputs but does not document a formal load test, concurrency test or response-time benchmark. The values below are proposed acceptance targets for a student demonstration deployment; they must be tested on the actual environment before being treated as measured results.

ID	Metric	Proposed Acceptance Target	Verification Method
NFR-PERF-01	Dashboard/API response	95% of ordinary read requests complete within 2 seconds under demo load.	Measure API latency over repeated requests.
NFR-PERF-02	Nearest-station calculation	Return pairing result within 1 second for up to 1,000 registered station records, excluding browser GPS acquisition time.	Seed test stations and measure service duration.
NFR-PERF-03	Concurrent users	Support 25 concurrent active browser sessions without failed requests under representative demo traffic.	Load test with scripted sessions.
NFR-PERF-04	Telemetry freshness	Dashboard reflects a completed simulation heartbeat within 5 seconds under normal demo conditions.	Compare server update timestamp with displayed update.
NFR-PERF-05	Canvas rendering	Maintain usable interaction at approximately 30 frames per second on a contemporary laptop for the standard scene.	Browser performance profiling on declared hardware.
NFR-PERF-06	Error handling	Invalid input and unauthorized requests return a controlled response rather than	Negative API tests.

		an unhandled server failure.	

3.4 Logical Database Requirements

The logical schema contains four principal entities. Telemetry records and environmental alerts reference a station through a foreign key. The user-to-station pairing is computed using geolocation and is not described as a stored foreign-key relationship in the supplied report.

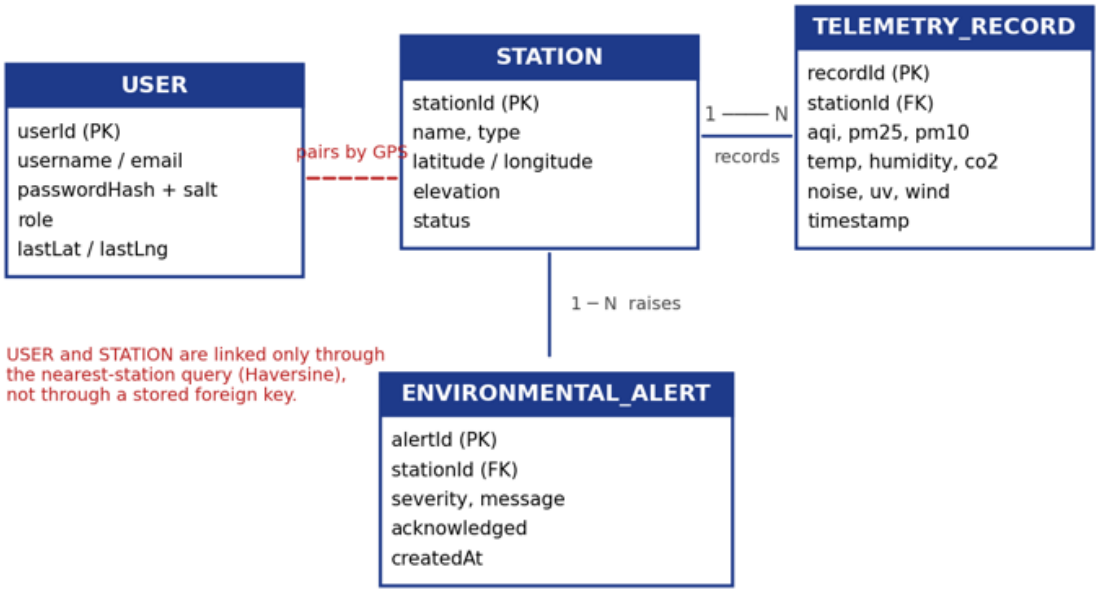


Figure 3.9: Entity–Relationship Diagram

Entity / Table	Key Fields	Constraints / Relationships
users	userId (PK), username, email, passwordHash, salt, role, lastLat, lastLng	Username and e-mail must be unique; passwordHash and salt required for authenticated accounts; role constrained to supported roles.
stations	stationId (PK), name, latitude, longitude, elevation, status	Coordinates within geographic ranges; referenced by telemetry and alerts.

telemetry_records	recordId (PK), stationId (FK), aqi, pm25, pm10, temperature, humidity, co2, noise, uv, wind, timestamp	Each record belongs to one station; one station may have many telemetry records.
environmental_alerts	alertId (PK), stationId (FK), severity, message, acknowledged, createdAt	Each alert belongs to one station; one station may have many alerts.

- The database shall preserve referential integrity between stations and associated telemetry/alerts.
- The application shall reject duplicate usernames or e-mail addresses at registration.
- The application shall store timestamps for telemetry and alert creation to support time-series views and report context.
- The application shall not store passwords in plaintext.
- If H2 fallback is used, the UI or operating documentation should warn that records are temporary and will be lost when the application stops.

3.5 Software System Attributes

3.5.1 Security

- NFR-SEC-01: Passwords shall never be stored in plaintext; the current prototype uses per-user salted SHA-256 as described in the project report.
- NFR-SEC-02: Protected endpoints shall validate authentication tokens and enforce role permissions for station registration, scenario injection and alert acknowledgement.
- NFR-SEC-03: Invalid credentials shall not result in a token being issued.
- NFR-SEC-04: Production deployment should use HTTPS, restrict CORS origins, protect secrets and replace fast SHA-256 password hashing with bcrypt or Argon2.
- NFR-SEC-05: Demonstration credentials documented in the report (admin/admin123 and operator/operator123) must be changed before any shared or production deployment.

3.5.2 Reliability

- NFR-REL-01: Invalid coordinates, missing stations, invalid scenarios and failed authentication shall be handled without application crash.

- NFR-REL-02: When MySQL is unavailable at startup, the documented fallback to H2 shall permit demonstration startup, while clearly indicating its non-persistent nature.
- NFR-REL-03: The system shall return an empty station result safely when no stations are registered.
- NFR-REL-04: Simulation and alert processing should not silently report success if the requested scenario or target alert is invalid.

3.5.3 Maintainability

- NFR-MNT-01: The application shall retain separation between presentation, controller, business logic and persistence layers.
- NFR-MNT-02: REST controllers, service logic, JPA entities and repositories should be independently testable.
- NFR-MNT-03: Endpoint names, scenario values, AQI rules and database schema shall be documented and version-controlled.
- NFR-MNT-04: Configuration such as port, database connection and CORS policy should be externalized rather than hard-coded.

3.5.4 Availability

- NFR-AVL-01: For an academic demonstration, the application should remain available during a scheduled 2-hour demonstration when the host and database are operational.
- NFR-AVL-02: A database outage at startup may trigger H2 fallback; loss of durable storage must be documented to the operator.
- NFR-AVL-03: The current system does not define high availability, multi-node clustering, automated disaster recovery or a production uptime SLA.

3.5.5 Usability and Accessibility

- NFR-USE-01: Citizen-facing output shall supplement numeric AQI with a plain-language verdict and advice.
- NFR-USE-02: Units shall be labelled and temperature unit changes shall be clear.
- NFR-USE-03: Permission-denied, empty-data, loading and API-error states shall be communicated to the user.

4. Appendices & References

Appendix A — Key Business Rules

Rule ID	Rule
BR-AQI-01	AQI 0–50 maps to “Crisp & Pristine”; advice indicates conditions ideal for outdoor activity.
BR-AQI-02	AQI 51–100 maps to “Moderate”; sensitive people should take care.
BR-AQI-03	AQI 101–200 maps to “Sensitive Warning”; limit outdoor exercise and use masks if needed.
BR-AQI-04	AQI above 200 maps to “Hazardous”; advise staying indoors and keeping windows closed.
BR-SCORE-01	Clean Air Score = round (100 – bounded AQI/5), where bounded AQI is clamped to 0…500.
BR-GEO-01	Valid latitude is −90…+90 and valid longitude is −180…+180.
BR-GEO-02	Nearest station is the station with minimum Haversine distance; bearing is normalized to 0…360 degrees.
BR-SIM-01	Accepted scenario values are HEATWAVE, INDUSTRIAL_EMISSION, RAIN_CLEANSING and NORMAL.
BR-DB-01	Telemetry and alert records reference a station; user-station proximity is calculated rather than represented as a stored foreign key.

Appendix B — Edge Cases and Expected Behaviour

The AETHERIS system is designed to handle invalid inputs and exceptional conditions safely. The following cases describe the expected system behaviour:

- **Invalid Geographic Coordinates:** If latitude is outside −90° to +90° or longitude is outside −180° to +180°, the system shall reject the coordinates and prevent distance calculation.
- **Location Permission Denied**: If the user denies browser location access, the system shall use a configured state-wise preset location.

- **No Registered Stations:** If no sensor stations are available, the system shall return an empty nearest-station result without causing an application failure.
- **AQI Exceeds 500:** The system shall limit the AQI value to 500, resulting in a Clean Air Score of 0 and a Hazardous verdict.
- **Invalid Login Credentials:** If the username or password is incorrect, the system shall reject the login request and shall not issue an authentication token.
- **Duplicate Registration:** If a username or e-mail address already exists, the system shall reject the registration request.
- **Unauthorized Operations:** If an ADMIN or OPERATOR attempts a protected operation without a valid authentication token, the system shall deny the request.
- **MySQL Unavailable:** If MySQL is unavailable during startup, the application shall fall back to the in-memory H2 database. The stored data will be temporary and may be lost when the application stops.
- **Unsupported Hazard Scenario:** If an unrecognized scenario is submitted, the system shall reject the request without applying any simulation effects.

Appendix C — Source-Report Verification Cases

Verification Case	Reported Expected / Observed Result	Status in Source Report
Haversine Jaipur to Delhi	235.29 km, bearing 36.15° (NE)	Passed
Same coordinate twice	0.00 km	Passed
0.001° latitude shift	Approximately 111.19 m	Passed
AQI boundary checks at 50, 51, 100, 101, 200, 201	Verdicts match defined threshold mapping	Passed
Clean Air Score at AQI 25, 150, 500	95, 70, 0	Passed
Temperature conversion at 0°C, 37°C, 100°C	32°F, 98.6°F, 212°F	Passed
Same password with different random salts	Different 64-character hashes; correct password verifies and wrong password fails	Passed

Appendix D — Requirement Traceability Overview

Feature Area	Requirement IDs	Verification Evidence
Authentication and roles	FR-AUTH-01 to FR-AUTH-04; NFR-SEC-*	Registration/login tests; invalid credential test; role authorization tests.
Station and GPS pairing	FR-GEO-01 to FR-GEO-04	Coordinate boundary tests; nearest-station distance/bearing tests; permission-denied path.
Telemetry and dashboard	FR-TEL-01 to FR-TEL-03; FR-DASH-01 to FR-DASH-04	API response checks; score/verdict boundaries; unit conversion; Canvas UI inspection.
Simulation and alerts	FR-SIM-01 to FR-SIM-02; FR-ALERT-01 to FR-ALERT-03	Scenario validation; role checks; alert threshold and acknowledgement tests.
Reporting and guided tour	FR-REP-01; FR-TOUR-01	Report print/PDF inspection; tour navigation test.
Performance and quality	NFR-PERF-*; NFR-REL-*; NFR-MNT-*; NFR-AVL-*	Load test, response-time sampling, negative tests and deployment observation.

Appendix E — Deployment and Operational Notes

- Run with Java 17 and the project Maven Wrapper as documented in the project repository.
- Default server port is 9090 unless changed in application.properties.
- Configure MySQL for durable data; treat H2 fallback as a temporary demo mode.
- Change all demonstration credentials before shared deployment.
- Use HTTPS and production-appropriate password hashing before deployment beyond the academic environment.
- Identify simulated telemetry clearly; integrate physical sensors only through a defined and tested station/gateway interface.

5. References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Spring Boot Reference Documentation: https://docs.spring.io/spring-boot/
- Spring Data JPA: https://spring.io/projects/spring-data-jpa
- Java SE 17 Message Digest API: https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/security/MessageDigest.html
- NIST FIPS PUB 180-4, Secure Hash Standard: https://csrc.nist.gov/publications/detail/fips/180/4/final
- MySQL Reference Manual: https://dev.mysql.com/doc/
- MDN Canvas API and Geolocation API: https://developer.mozilla.org/
- U.S. EPA / AirNow AQI information: https://www.airnow.gov/aqi/
- Sinnott, R. W., "Virtues of the Haversine," Sky and Telescope, vol. 68, no. 2, 1984.
- AETHERIS Project Report supplied with this request: architecture, algorithm, API, database, tests and limitations.

Approval / Review

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